PROMOFINDER AI — TECHNICAL ARCHITECTURE & SOLUTION WHITEPAPER

AUTOMATED PROMO DISCOVERY, SCREENPLAY GROUNDING & DETERMINISTIC VALIDATION

1. Strategic Business Framing & Leverage (Why Problem 1?)

In vertical microdramas (1–2 minute episodes), audience acquisition is overwhelmingly driven by paid performance marketing (Meta Reels, TikTok Spark Ads, YouTube Shorts). Conversion curves are brutally steep: **~65% of scrollers drop off within 3 seconds**, and over 80% abandon by second 10. Currently, human editors spend 30–45 minutes per episode manually scrubbing footage to identify promo cuts. Across a 50-episode show, this demands **25 to 37.5 hours of senior editorial labor**, creating a severe operational bottleneck. **PromoFinder AI** automates this pipeline directly from pre-production screenplays, shifting editors from manual hunting to high-value curation. Surfacing the top 5% highest-retention cuts reduces turnaround from days to seconds, allowing marketing teams to test 10x more ad creative variants and significantly reducing Blended Customer Acquisition Cost (CAC).

2. Operational Assumptions & Screenplay Domain Realities

Dimension	Screenplay Domain Reality	Engineering Rationale & System Implementation
Clock Coordinate Space	Scene-relative clock resets (every scene starts at 00:00.00).	Verified across all 50 episodes. Storing naive global timestamps would cause fatal multi-scene desync. All segments maintain an independent (episode, scene, start_ts, end_ts) coordinate tuple.
Duration Sweet Spot	Hard constraint: [20s, 60s]. Social ad completion peaks at 28s–42s.	Implemented a continuous polynomial decay curve centered at 35.0s: 10.0 - 3.2×(|d - 35|/25)^1.4. Avoids discrete step-function cliffs; clips <20s or >60s are strictly disqualified.
Producer SFX Cues	Script tags like CLIFFHANGER SFX are authorial hints, not verdicts.	Writers insert stings over mundane dialogue. SFX is treated as a secondary calibration signal; genuine unresolved dialogue tension is required for high scores.
Hallucination Safety	LLM summaries are interpretive prose; dialogue must be verbatim.	Deterministic regex substring verification checks quoted lines ("...") against the raw screenplay text with ±0.5s tolerance, guaranteeing zero invented dialogue lines.

3. System Architecture & Component Separation (~65% Deterministic / ~35% LLM)

The architecture enforces a strict boundary between deterministic procedural code and qualitative LLM reasoning:

• **Parser (parser.py):** State-machine regex parser extracting scenes, character dialogue, emotions, and scene-relative timestamps. Handles both standard and Episode-6 ALL-CAPS formats, plus inline bracket syntax.

• **Windowing Engine (analyzer.py):** Generates heuristic candidate cuts per scene (Full Scene, First-45s Hook, Climax Cut, SFX-Anchored). Produces an average of 4.3 candidates per episode.

• **Pluggable LLM Client (llm_client.py):** Standardized interface supporting Groq (LLaMA 3.3 70B), OpenAI (GPT-4o-mini), Google Gemini, or grounded Mock, backed by a content-addressed SHA-256 disk cache (outputs/cache/).

• **Deterministic Validator (validator.py):** 100% Python enforcement—duration bounds [20s, 60s], timestamp existence (±0.5s), dialogue grounding, and temporal graph deduplication.

• **Cloud Deployment:** Vercel SPA (React 18/Tailwind CSS) ↔ Vercel Edge Reverse Proxy (/api/*) ↔ Azure Basic B1 FastAPI Backend ↔ Supabase Managed PostgreSQL. Screenplay source viewer streams all 50 full scripts via CDN episodes.json.

4. Explainable 5-Dimension Grounded Scoring Rubric

Dimension	Weight	Engine	Screenplay Grounding & Evaluation Formula
Hook Strength	25%	AI + Grounded	First 3s trigger: dialogue onset speed (≤1.5s gives +0.9), emotional directions (Cold, Defiant, Whispering), line brevity (15–95 chars).
Cliffhanger Strength	30%	AI + Grounded	Unresolved tension at cut: terminal punctuation (?, !, ..., —), climax cliffhanger SFX within final 10s, abrupt conversation cutoff.
Standalone Clarity	20%	AI + Grounded	Context intelligibility for cold traffic: dialogue density (≥0.55), two-speaker duets (scenes with >3 active speakers receive context clutter penalty).
Emotional Impact	15%	AI + Grounded	Visceral stakes: density of high-conflict domain keywords (*betrayal, marriage, scars, murder, duke, honor, blade, fire, dead*).
Duration Quality	10%	Deterministic	Continuous mathematical decay: 10.0 - 3.2×(|duration - 35.0| / 25.0)^1.4. 35.0s → 10.0; 25s/45s → 9.36; 20s/60s → 7.20.

PROMOFINDER AI — TELEMETRY, COST MODELING & FAILURE AUDIT

GRAPH DEDUPLICATION MATH, TELEMETRY FUNNEL, UNIT ECONOMICS & AI FAILURE REFLECTION

5. Deduplication Algorithm & Graph Theory Mathematics

Candidate windowing frequently creates overlapping cuts (e.g., a 45s opening hook vs. a 55s full scene of the same dialogue). In naive systems, teams confuse pairwise collisions with filtered candidates. We formulate this rigorously using graph theory:

1. Construct an **Intersection Graph** $G = (V, E)$, where vertices V represent candidate cuts ($|V| = 218$). An edge $e = (u, v) \in E$ exists if the temporal Intersection-over-Union (IoU) of dialogue lines between candidates u and v exceeds **60%**: $\text{IoU}(u, v) = |u \cap v| / |u \cup v| \geq 0.60$.
2. Across 50 episodes, this detects exactly **133 pairwise collision edges**.
3. A **Greedy Maximal Independent Set** algorithm sorts candidates by preliminary score. When the highest-scoring candidate is selected, all connected adjacent vertices are eliminated together.
4. **Result:** Exactly **96 redundant candidate vertices are pruned**, leaving **122 distinct, non-overlapping candidate clips** in the ranking pool.

6. Telemetry Funnel & Cost Modeling at Enterprise Scale

Empirical 50-Episode Pack Telemetry: 218 Total Evaluated → 122 Passed Validation (**56.0% pass rate**) → 96 Pruned as Redundant (**44.0% prune rate**) → 122/122 Accepted Pool Constraint Compliance (**100.0%**: 0 timestamp errors, 0 hallucinations, 0 duration violations) → Top 20 Ranked Output (Scores: 7.591 to 8.534, Mean: 7.914).
Rank #1 Promo (Ep 36, 44.8s, Score: 8.534): Hook: *ALEXANDER: (Simmering with Fury) "They are marrying in two days..."* ↔ Cliffhanger: *MOTHER: "Then why did they leave..."*

Pipeline Execution Tier	Cost per Episode	Cost per 50-Ep Show	End-to-End Latency	Commercial Operational Viability
Groq (LLaMA 3.3 70B)	**$0.0005** (~650 tokens)	**$0.025** (2.5 cents)	14.2 seconds	Ideal for high-volume batch processing; ultra-low inference cost.
OpenAI (GPT-4o-mini)	**$0.0021** (~650 tokens)	**$0.105** (10.5 cents)	32.5 seconds	Standard enterprise fallback; exceptional JSON structured output adherence.
Manual Human Editor	~$25.00 (35 mins labor)	~$1,250.00 (30 hours)	3 to 5 business days	Severe bottleneck; inconsistent creative quality and high human fatigue.

7. Transparent System Limitations & Engineering Edge Cases

Limitation	System Impact	Engineering Mitigation Strategy
Sub-20s Short Scenes	Ep 39 & Ep 44 produce 0 promo windows.	Intended design: clips under 20s lack sufficient narrative body for social ad retention.
Audio Delivery Blindness	Text scripts omit vocal cadence & pause timings.	Parsed emotional stage directions (Coldly, Whispering, Terrified) act as weighted proxies.
Crowd Cross-Talk Density	Scenes with 4+ speakers degrade standalone clarity.	Standalone clarity metric penalizes scenes exceeding 3 active speakers to favor focused 2-person duets.

8. AI Tools Used & Documented AI Development Failure

AI Tools Utilized: Google Gemini 1.5 Flash API, OpenAI GPT-4o-mini, Groq LLaMA 3.3 70B, and Antigravity Agent for iterative architectural review.
Documented AI Failure During Development: Early in pipeline construction, an AI-generated instruction prompted the hallucination validator to check whether the *entire candidate rationale sentence* was a verbatim substring of the screenplay text. This created an immediate, catastrophic false-positive failure: **100% of generated candidates (218/218) were disqualified as hallucinations**, because LLM rationales are interpretive prose (e.g., *"Alexander confronts the matriarch regarding his scars"*), which never appears literally in scripts.
The Fix: We re-engineered the validator in deterministic Python using regular expressions to isolate only text enclosed in quotation marks ("...") with ≥10 characters, validating only dialogue quotes against the source text. This restored 100% validation fidelity while preventing real hallucinations.

9. Zero-Modification Scaling to Future Shows & Production Ingestion

PromoFinder AI requires **zero source code modifications** to process any upcoming Kuku FM series: placing raw .txt files into data/scripts/<show_name> and executing python src/pipeline.py automatically detects scene markers, headings, timestamp bounds, and formatting variants. For live ad teams, the deployed web application features an on-demand **Pipeline & Ingestion Tab** allowing editors to paste raw script text or enter public URLs (Google Drive, Dropbox, S3) to discover and rank promo moments in real time via the Azure backend.